

4.1

Answer each of the following

- Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- Suppose that a simple linear regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- Suppose that a simple linear regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\hat{e}_i^2 = 182.85$, find R^2 .

Ans:

- Note that $SST = \sum (y_i - \bar{y})^2 = (\sum y_i^2) - N\bar{y}^2 = 7825.94 - 20 * 19.21^2 = 445.458$. Therefore, by definition of R^2 , $R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} = 0.8429$.
- $\hat{\sigma}^2 = \frac{\sum (y_i - \hat{y}_i)^2}{N-2} = \frac{SSE}{N-2} = \frac{SST-SSR}{N-2} = \frac{SST \cdot (1-R^2)}{N-2} = \frac{725.94 \cdot (1-0.7911)}{18} = 8.4249$
- $R^2 = \frac{SSR}{SST} = \frac{SST-SSE}{SST} = \frac{[\sum (y_i - \bar{y})^2] - \hat{e}_i^2}{\sum (y_i - \bar{y})^2} = \frac{631.63 - 182.85}{631.63} = 0.7105$

4.3

We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator.

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

Ans:

- $\hat{y}_i = 1.2 + 0.8 \times 4 = 4.4$
- Note that $\bar{x} = \frac{3+2+1-1+0}{5} = 1$ and $\hat{\sigma}^2 = \frac{\sum_{i=1}^5 (y_i - \hat{y}_i)^2}{N-2} = \frac{\hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$. Therefore, we have $se(f) = \sqrt{var(f)} = \sqrt{\hat{\sigma}^2(1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2})} = \sqrt{1.2(1 + \frac{1}{5} + \frac{(4-1)^2}{10})} = \sqrt{1.2 \cdot 2.1} = 1.5875$
- The $(1 - \alpha)\%$ confidence interval with $df = 3$ is

$$[\hat{y}_i - t_{\frac{\alpha}{2}}(3) \times se(f), \hat{y}_i + t_{\frac{\alpha}{2}}(3) \times se(f)] \quad (1)$$

Since $t_{0.025}(3) = 3.182$. Therefore, 95% prediction interval for y given $x_0 = 4$ is $[4.4 - 3.182 \times 1.5875, 4.4 + 3.182 \times 1.5875] = [-0.651425, 9.451425]$

- Since $t_{0.005}(3) = 5.841$ By (1), 99% prediction interval for y given $x_0 = 4$ is $[4.4 - 5.841 \times 1.5875, 4.4 + 5.841 \times 1.5875] = [-4.8725875, 13.6725875]$
- Note that $se(f) = \sqrt{var(f)} = \sqrt{\hat{\sigma}^2(1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2})} = \sqrt{1.2(1 + \frac{1}{5} + 0)} = 1.2$ and $t_{0.025}(3) = 3.182$. By (1), 95% prediction interval for y given $x_0 = \bar{x} = 1$ is $[1.2 + 0.8 \times 1 - 3.182 \times 1.2, 1.2 + 0.8 \times 1 + 3.182 \times 1.2] = [-1.8184, 5.8184]$ Compare with (c), since the degree of freedom and confidence level are the same, and the standard error in (e) is smaller than (c), the width of the confidence interval are narrower than (c).

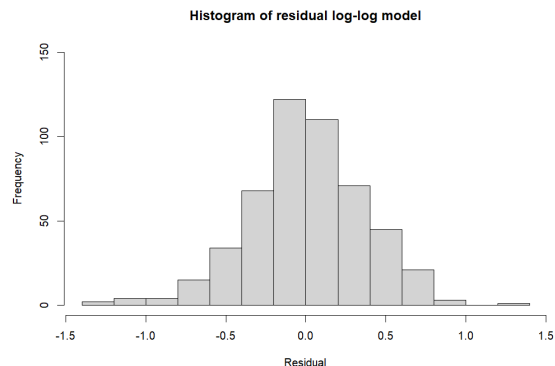
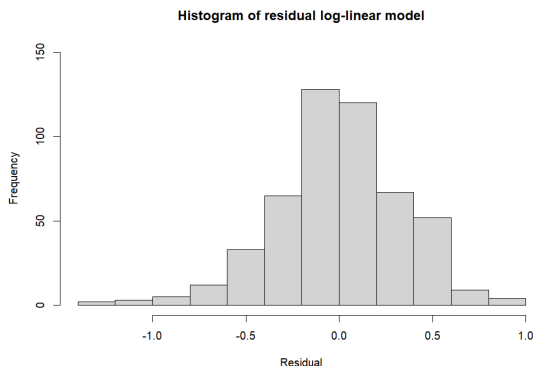
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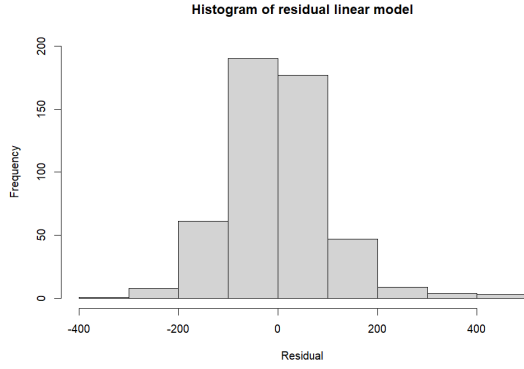
The file `collegetown` contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file `collegetown.def`.

- Estimate the log-linear model $\ln(\text{PRICE}) = \beta_1 + \beta_2 \text{SQFT} + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Estimate the log-log model $\ln(\text{PRICE}) = \alpha_1 + \alpha_2 \ln(\text{SQFT}) + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Compare the R^2 value from the linear model $\text{PRICE} = \delta_1 + \delta_2 \text{SQFT} + e$ to the “generalized” R^2 measure for the models in (b) and (c).
- Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- For each of the models in (a)–(c), plot the least squares residuals against SQFT. Do you observe any patterns?
- For each model in (a)–(c), predict the value of a house with 2700 square feet.
- For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

Ans:

- Running the code, the regression can be expressed as $\ln(\text{Price}) = 4.393866 + 0.03604449 \cdot \text{SQFT} + e$. Since $\text{PRICE} = \exp(\beta_1 + \beta_2 \text{SQFT})$. The slope is $\frac{d\text{PRICE}}{d\text{SQFT}} = \exp(\beta_1 + \beta_2 \text{SQFT}) \cdot \beta_2 = 7.800865$. The elasticity is $\frac{d\text{PRICE}}{d\text{SQFT}} \frac{\text{SQFT}}{\text{PRICE}} = \beta_2 \text{SQFT} = 0.9833701$.
- Running the code, the regression can be expressed as $\ln(\text{Price}) = 2.049653 + 1.024828 \cdot \ln(\text{SQFT}) + e$. Since $\text{PRICE} = \exp(\alpha_1 + \alpha_2 \ln(\text{SQFT})) = e^{\alpha_1} \text{SQFT}^{\alpha_2}$. The slope is $\frac{d\text{PRICE}}{d\text{SQFT}} = \alpha_2 e^{\alpha_1} \text{SQFT}^{\alpha_2 - 1} = 8.638821$. The elasticity is $\frac{d\text{PRICE}}{d\text{SQFT}} \frac{\text{SQFT}}{\text{PRICE}} = \alpha_2 = 1.024828$.
- Running the code, R^2 of the linear model is 0.6413167. Since the generalized R^2 is defined as $R_g^2 = r_{y, \hat{y}}^2$, the generalized R^2 of the log-log model is 0.6445084, and generalized R^2 of the linear model 0.6413167. The log-log model produces a slightly higher generalized R^2 (0.6445 vs 0.6413), indicating a marginally better fit.
- The histograms of residuals for three models are shown as below respectively.





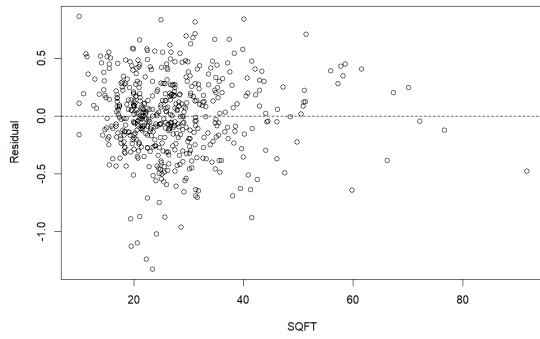
Jarque-Bera statistics of model (a) is $X^2 = 26.679$, $df = 2$, $p\text{-value} = 1.61e - 06$.

Jarque-Bera statistics of model (b) is $X^2 = 14.279$, $df = 2$, $p\text{-value} = 0.0007933$.

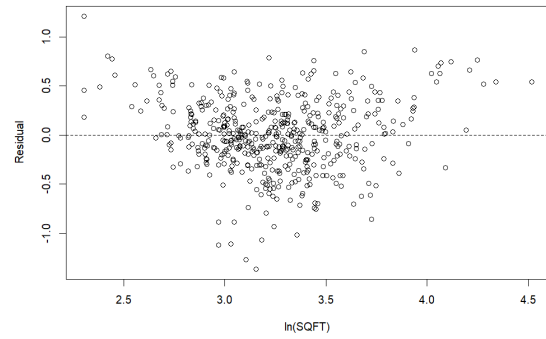
Jarque-Bera statistics of model (c) is $X^2 = 221.11$, $df = 2$, $p\text{-value} < 2.2e - 16$.

Based on these results, the residuals of the log-log model appear closest to normality, as it has the smallest Jarque-Bera statistic.

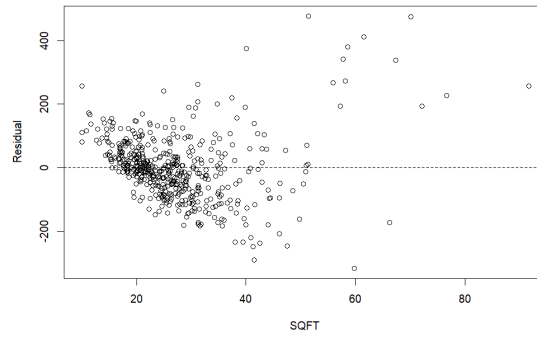
(e) The least squares residuals against SQFT for three models are shown as below respectively.



(a) Log-Linear model



(b) Log-Log model



(c) Linear model

For the log-linear model, the residuals appear more randomly scattered around zero, although there is still slight variation in spread.

The the log-log model shows the most random pattern, with residuals evenly distributed around zero and no clear systematic structure. The variance of residuals also appears more stable.

For the linear model, the spread of residuals increases for larger values of SQFT, indicating possible heteroskedasticity.

Overall, the residual plots suggest that the log-log model provides the best fit among the three models.

(f) With 2700 square feet ($x = 27$):

In model (a) the predicted value is $PRICE = \exp(\beta_1 + \beta_2 \cdot 27) = 214.2336$ (in thousands).

In model (b) the predicted value is $PRICE = \exp(\alpha_1 + \alpha_2 \ln(27)) = 227.5386$.

In model (c) the predicted value is $PRICE = \delta_1 + \delta_2 \cdot 27 = 246.4557$.

(g) The $(1 - \alpha)\%$ prediction interval with $df = 498$ is

$$[\hat{y}_i - t_{\frac{\alpha}{2}}(498) \times se(f), \hat{y}_i + t_{\frac{\alpha}{2}}(498) \times se(f)]$$

, where $se(f) = \sqrt{\hat{\sigma}^2(1 + \frac{1}{N}) + (x_0 - \bar{x})^2 var(b_2)}$

For log-linear model, $x = 27$ and $\bar{x}$ is the sample mean of SQFT. The prediction interval calculated above is the interval of $\ln(y)$, we need to take exponential for both bounds. Therefore, the prediction interval is [109.7685, 418.1167].

For log-log model, $x = \ln(27)$ and $\bar{x}$ is the sample mean of $\ln(sqft)$. The prediction interval calculated above is the interval of $\ln(y)$, we need to take exponential for both bounds. Therefore, the prediction interval is [111.1406, 465.8406].

For linear model, $x = 27$ and $\bar{x}$ is the sample mean of SQFT. The prediction interval is [44.27727, 448.6341].

(h) The log-log model is preferred because it has the highest generalized R^2 , the smallest Jarque–Bera statistic, and the most randomly distributed residuals.